CLOUD COMPUTING

ASSIGNMENT 01

Git Repository Management, Git LFS & GitHub Pages Portfolio

STUDENT INFORMATION

Details	Information
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Course Code	BSE-410
Subject	Cloud Computing
Assignment	Assignment 01

1. INTRODUCTION

This assignment demonstrates practical cloud computing and version control skills using Gitea, GitHub, Git Large File Storage (LFS), and GitHub Pages. It covers repository hosting, managing multiple Git remotes, handling large files, and deploying a personal portfolio website.

2. TASK 1 — GITEA SETUP AND REPOSITORY PUSH

Objective

To configure Gitea on an Ubuntu server and push a local Git repository to the Gitea remote repository.

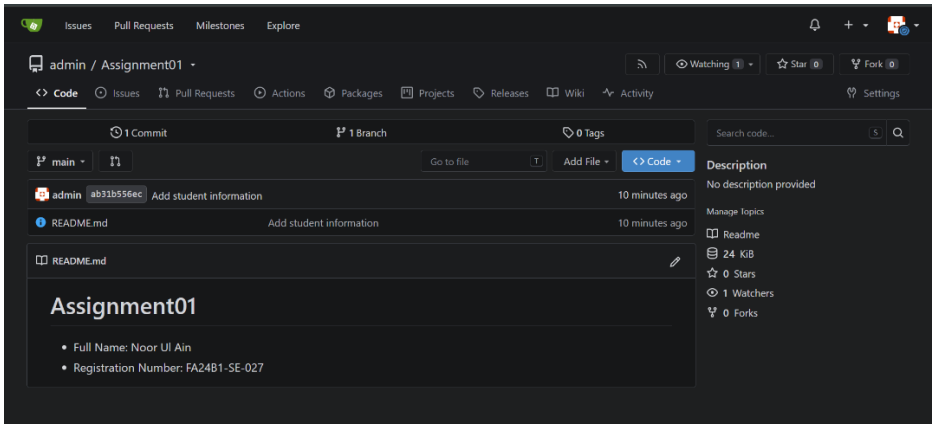
Description

Gitea was used as a self-hosted Git repository management platform. A repository named **Assignment01** was created, and the local project was pushed to the Gitea server.

Screenshot Evidence

```
noorulain027@ubuntu:~/Gitea$ cd ~/Gitea
noorulain027@ubuntu:~/Gitea$ docker compose ps
NAME       IMAGE                COMMAND                  SERVICE
   CREATED         STATUS         PORTS
gitea      gitea/gitea:latest   "/usr/bin/entrypoint…"   gitea
   8 minutes ago   Up 8 minutes   22/tcp, 0.0.0.0:3000->3000/tcp
, [::]:3000->3000/tcp
gitea_db   postgres:17-alpine   "docker-entrypoint.s…"   db
   8 minutes ago   Up 8 minutes   5432/tcp
noorulain027@ubuntu:~/Gitea$ curl -sS -o /dev/null -w "Gitea HTT
P status: %{http_code}\n" http://127.0.0.1:3000/
Gitea HTTP status: 200
noorulain027@ubuntu:~/Gitea$
```

- task1_gitea_running.png



- task1_gitea_push.png

```
noorulain027@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://admin@127.0.0.1:3000':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 299 bytes | 149.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To http://127.0.0.1:3000/admin/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'gitea/main'.
noorulain027@ubuntu:~/Assignment01$ git remote -v
gitea   http://admin@127.0.0.1:3000/admin/Assignment01.git (fetc
h)
gitea   http://admin@127.0.0.1:3000/admin/Assignment01.git (push
)
noorulain027@ubuntu:~/Assignment01$
```

- task1_gitea_repository.png

3. TASK 2 — GITHUB AS A SECOND REMOTE

Objective

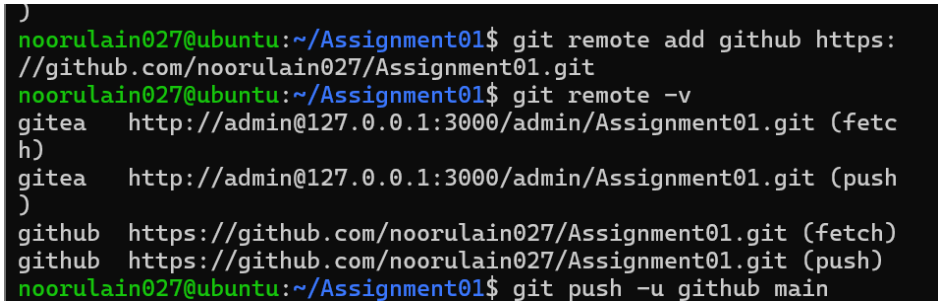
To configure GitHub as an additional remote repository and push the project to GitHub.

Description

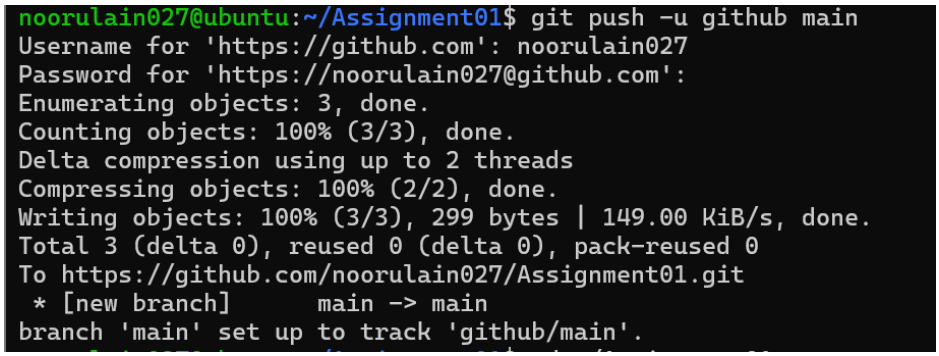
A second remote was configured so the project could be managed through both Gitea and GitHub. The configured remotes were checked, and the project was pushed to GitHub.

Screenshot Evidence

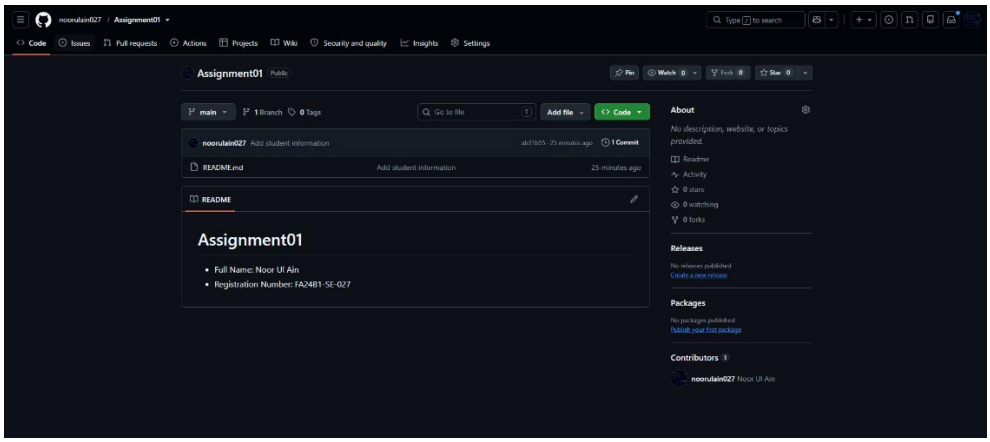
- `task2_remotes.png`



- `task2_github_push.png`



- `task2_github_repository.png`



GitHub Repository:
https://github.com/noorulain027/Assignment01

4. TASK 3 — GIT LARGE FILE STORAGE (LFS)

Objective

To configure Git LFS for managing large binary files in a Git repository.

Description

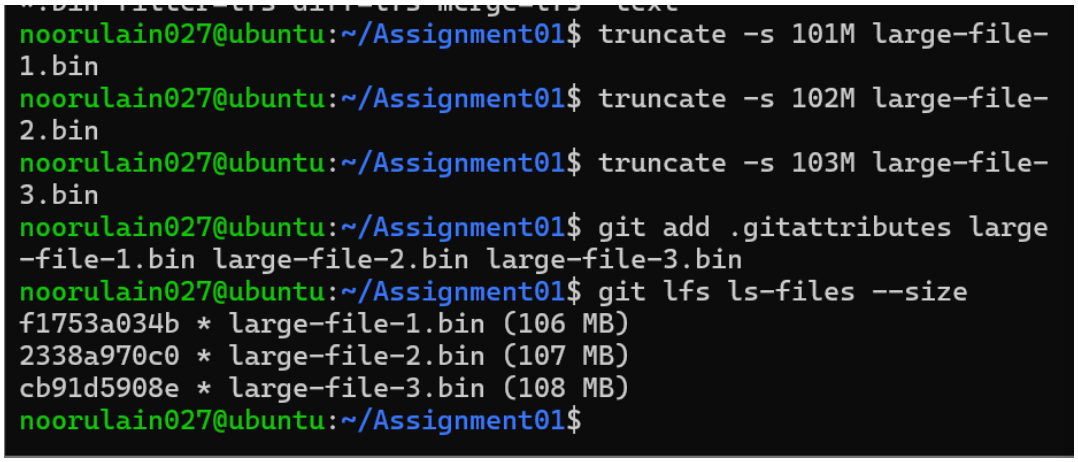
Git Large File Storage (LFS) was configured to track `.bin` files. Three large binary files were created, added to the repository, committed, and prepared for pushing to GitHub using Git LFS.

Screenshot Evidence

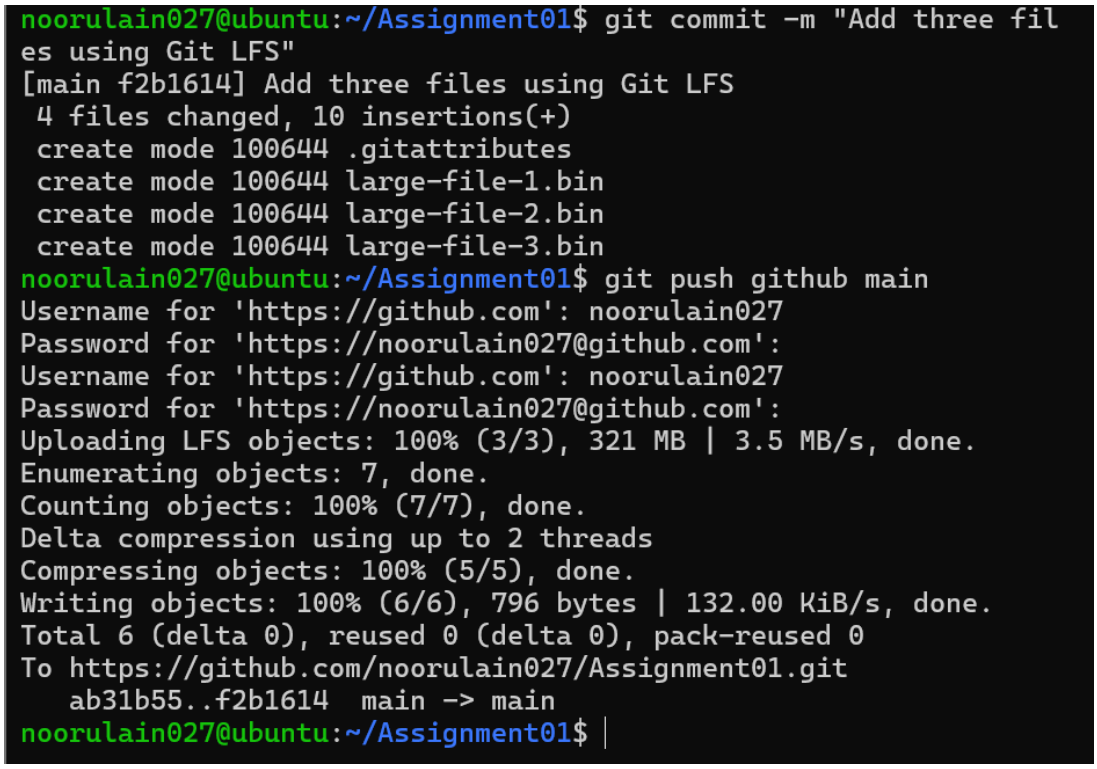
- `task3_lfs_setup.png`

```
noorulain027@ubuntu:~/Assignment01$ cd ~/Assignment01
noorulain027@ubuntu:~/Assignment01$ git lfs install
Updated Git hooks.
Git LFS initialized.
noorulain027@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.4.1 (GitHub; linux amd64; go 1.22.2)
noorulain027@ubuntu:~/Assignment01$ git lfs track "*.bin"
Tracking "*.bin"
noorulain027@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
noorulain027@ubuntu:~/Assignment01$
```

- task3_lfs_files.png



- task3_lfs_push.png



5. TASK 4 — GITHUB PAGES PORTFOLIO

Objective

To create and publish a personal portfolio website using GitHub Pages.

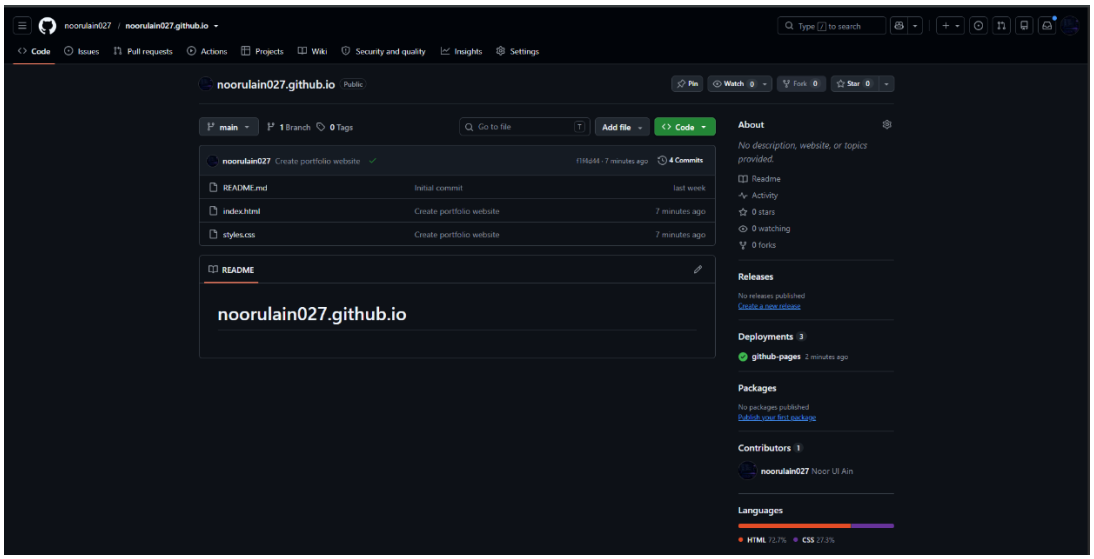
Description

A personal portfolio website was developed using HTML and CSS. The website presents personal information, education, skills, and projects. GitHub Pages was used to publish the website online.

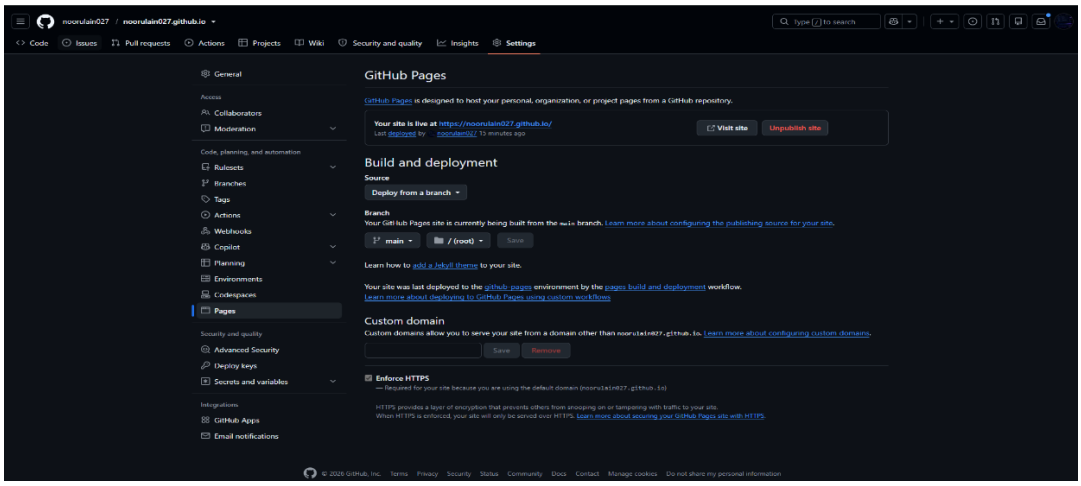
Live Portfolio:
https://noorulain027.github.io/

Screenshot Evidence

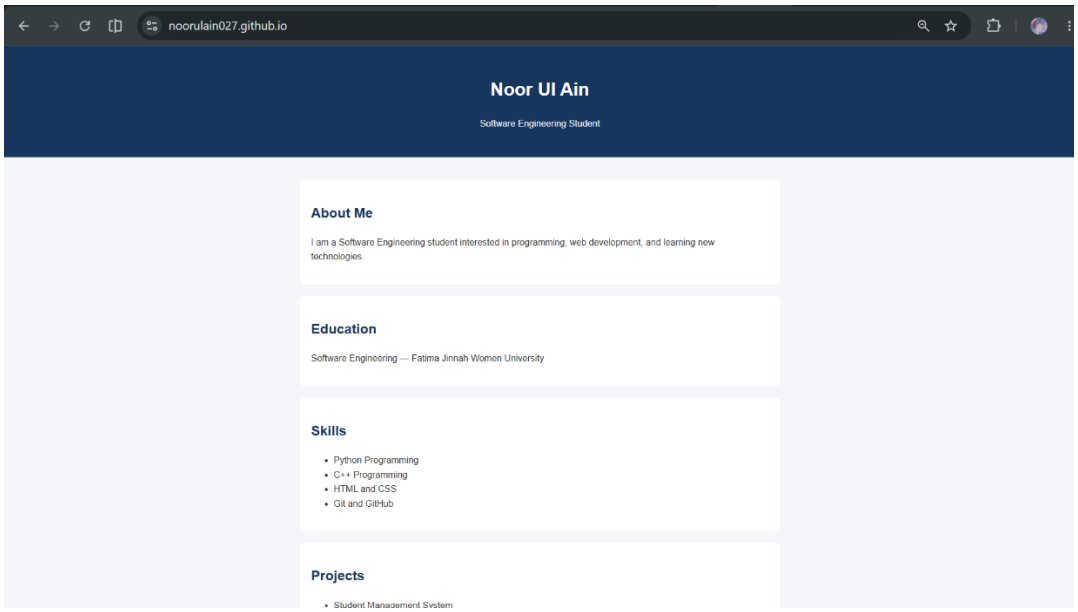
- `task4_pages_repository.png`



- `task4_pages_deployment.png`



- `task4_portfolio_live.png`



6. CONCLUSION

This assignment provided practical experience with Git repository hosting, multiple remote repositories, Git Large File Storage, and website deployment through GitHub Pages. It demonstrated how development files can be managed using Git and hosted online using cloud-based services.

The included screenshots serve as evidence of the practical tasks performed.

END OF ASSIGNMENT

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